

Project Title: UET CodeRank

Departmental LeetCode Leaderboard and Stats Monitoring
System

Member details:

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Description

UET CodeRank is a desktop application built for the Computer Science Department of UET Lahore. It allows students to register with their LeetCode username and join a batch and section. The system periodically fetches their LeetCode statistics through an external API and displays a department wide leaderboard ranked by problems solved. Admins manage batches, sections, enrollment approvals, and can generate reports on student performance.

Features/Modules

1. Admin Control Panel (User & Content Management)
2. Student Registration & Profile Tracking
3. Automated LeetCode Data Synchronization
4. Real-time Leaderboard System (Global/Batch/Section)
5. Badge & Notification System
6. Reporting & Logging Module

Use Cases

Use case ID	Use Case Name	Interface Id
U1.1	Admin Login	I1.1
U1.2	Add/Edit/Delete Batch	I1.2
U1.3	Add/Edit/Delete Section	I1.2
U1.4	View Enrollment Requests	I1.3
U1.5	Approve/Reject Enrollment	I1.3
U1.6	Sync Student Stats	I1.4
U1.7	View Leaderboards (Overall/Batch)	I1.5
U1.8	Award Badge Manually	I1.5
U1.9	Generate Reports	I1.6
U1.10	View Error Logs	I1.7
U1.11	Add/Delete Admin	I1.8
U2.1	Register Account	I2.1
U2.2	Student Login	I2.1
U2.3	Request to Join Section	I2.2
U2.4	View Profile & Stats	I2.3
U2.5	View Leaderboard	I2.4
U2.6	View Own Badges	I2.5
U2.7	View Notifications	I2.6
U2.8	Update Profile	I2.3

Detailed Use cases

Important use cases details:

U1.1 Admin Login

The admin provides their username and password. The system verifies credentials against the admin table. If successful then session is created and the admin is redirected to the dashboard. And if there are few wrong login attempts then it lock the login for few minutes and log to error log.

U1.2 Add/Edit/Delete Batch

Admin can create a new academic batch (e.g., "2025"). For editing, the admin updates the batch name. Deletion is only permitted if no students are currently assigned to that batch to maintain database integrity.

U1.5 Approve/Reject Enrollment

Admin reviews student join requests for specific sections. Admin can either approve (adding student to section) or reject (deleting request). This updates the enrollment requests table. Admin is also notified if the user entered a correct leetcode username.

U1.6 Sync Student Stats

The admin triggers the sync button and the system loops through all approved students. It calls the LeetCode Alfa API to fetch solved counts and contest ratings, then updates the leetcode stats table. The system also implement smart automatic sync system, in which using the last updated attribute of student it smartly fetch the data.

U1.9 Generate Reports

The admin selects a filter (Overall or Batch-wise). The system aggregates data and

generates a PDF document. The pdf document can be something like a summary for specific batch or students providing a complete insight into student performance in one PDF. **U2.1 Register Account**

The student fills out a form with their Name, Roll Number, and LeetCode Username. The system checks for duplicate roll numbers. Upon submission, the account is created with a PENDING status.

U2.3 Request to Join Section

Student selects their specific Batch and Section. A request is sent to the enrollment requests table. The student remains restricted until an Admin approves this request. A student can join only a single batch although it can join multiple

U2.5 View Leaderboard

The student accesses the ranked list of peers. The system calculates rank based on the "Total Solved" count. There will be some points according to hard, easy or medium solved problems and then students are ranked according to their points. Students can toggle views between their own Section, Batch or the Department side leaderboard. It promotes a sense of competition.

Interfaces

I1.1 Admin Login Screen

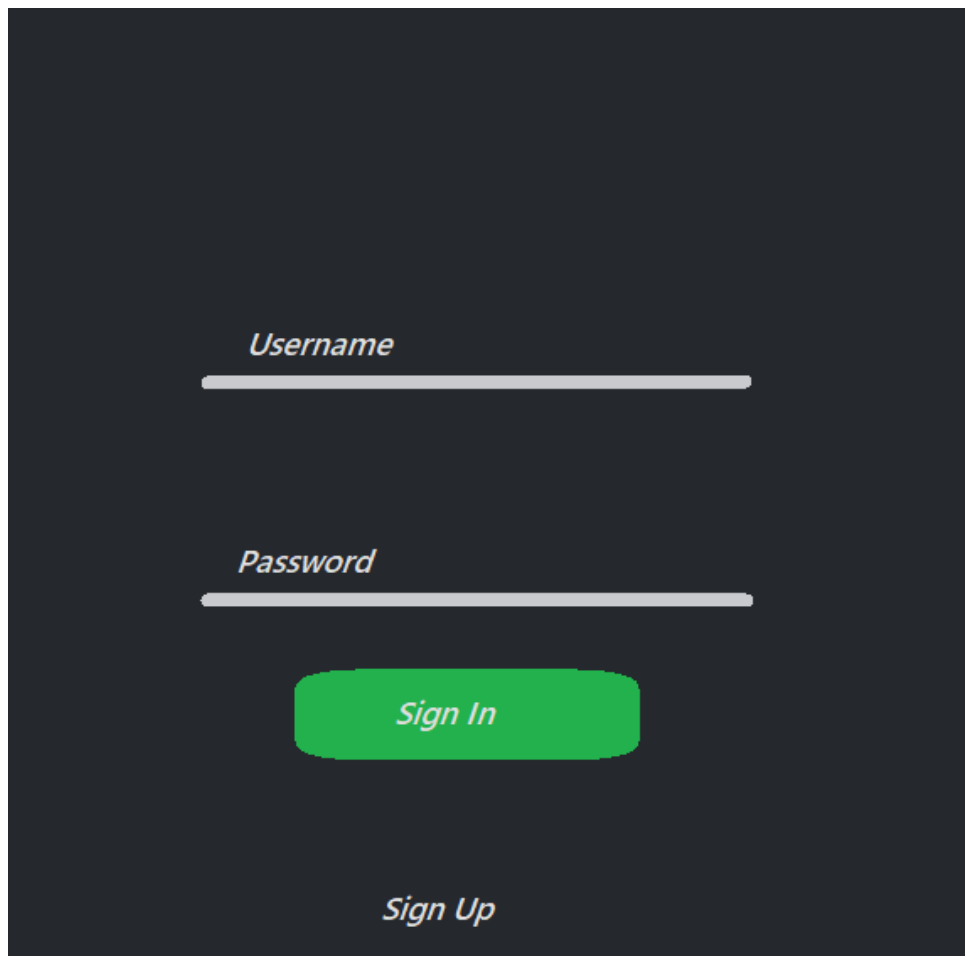


Figure 1: Sample UI for login

Validators List:

- Username must exist in Admin table.
- Password cannot be empty.
- Error message shown for incorrect credentials.

I2.1 Student Registration Screen

Validators List:

- Roll Number must follow YYYY-CS-XX format.
- LeetCode username must be valid (checked via API).
- Username must be unique.

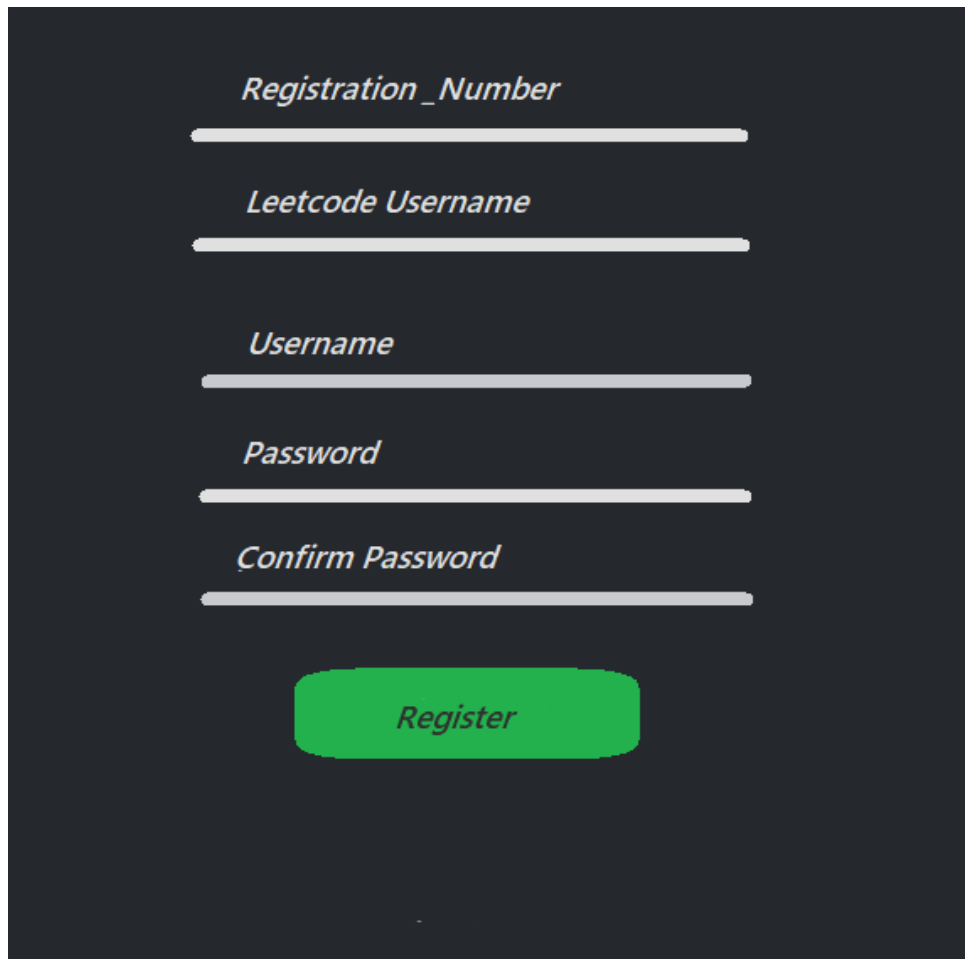
A sample user interface for student registration. It features a dark blue background with five white input fields stacked vertically. The labels for the fields are 'Registration_Number', 'Leetcode Username', 'Username', 'Password', and 'Confirm Password', all in a light blue, italicized font. Below the input fields is a red rounded rectangular button with the word 'Register' in a light blue, italicized font.

Figure 2: Sample UI for student registration

- Password and Confirm Password fields must match.

I2.3 Enrollment Requests Screen

Validators List:

- User can request more than one session.
- If user is already in another session then it cannot join another one, although we can still see available sessions.

I2.5 Leaderboard screen

Validators List:

- Batch and Section filters must load only active categories from the database.
- Search queries must be alphanumeric only.
- The UI must display a "No Data Found" message if a selected section has zero approved students.
- Must ensure the ranking algorithm correctly handles identical problem counts.

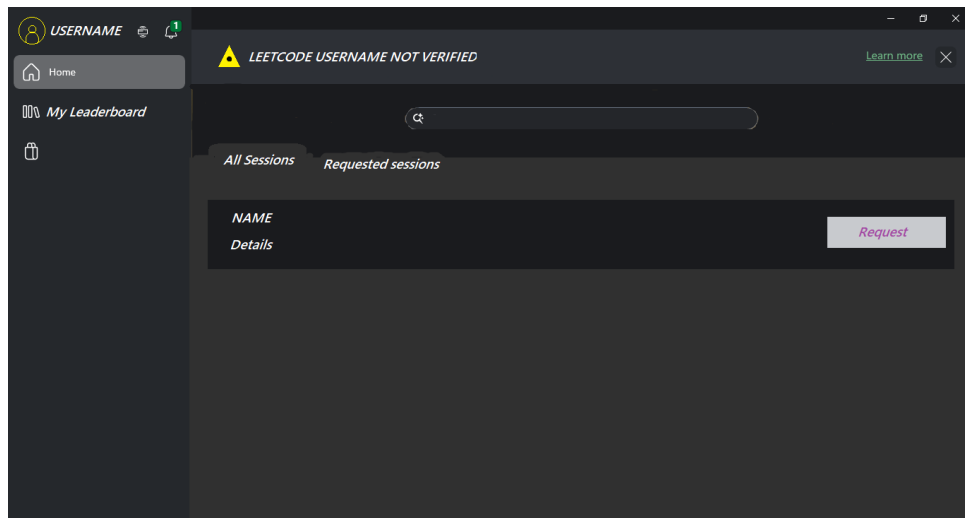


Figure 3: Sample UI for available sessions

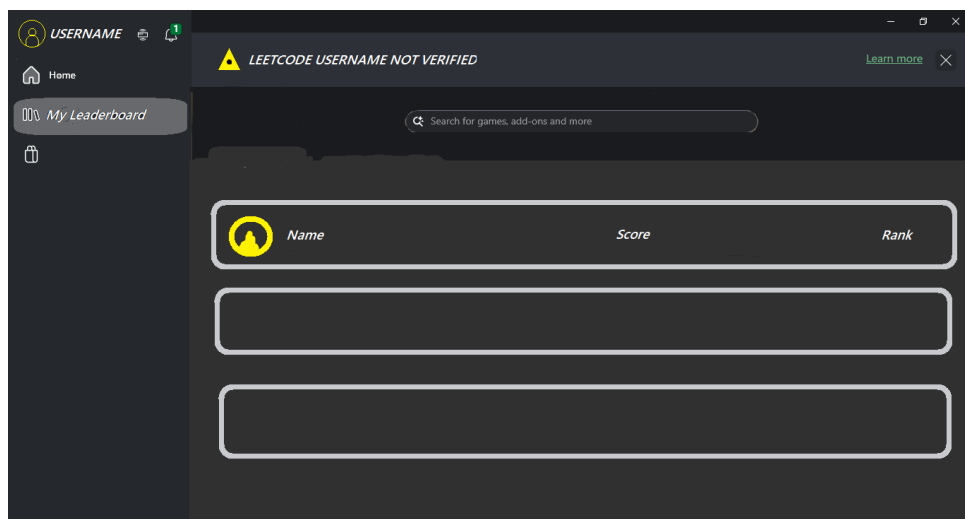


Figure 4: UI for leaderboard

GitLab Repository

<https://gitlab.com/umar342342342/UET-CodeRank>

Plan

Use CaseId	Assignee	Deadline
U1.1 – U1.10	Muhammad Umar Shahid	22-04-2026
U1.11 – U2.5	Taha Zulfiqar	25-04-2026
U2.6 – U2.8	Noor-Ul-Haq	02-05-2026